

Input:2

```
real    0m0.017s
user    0m0.010s
sys     0m0.006s
```

Input:10

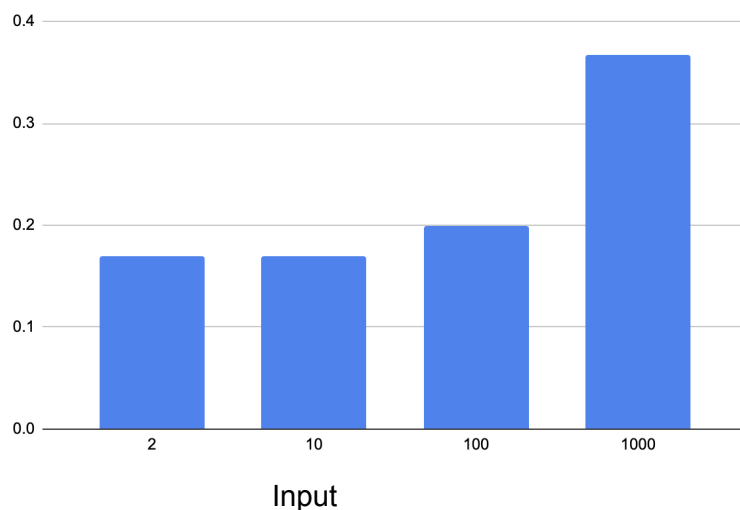
```
real    0m0.017s
user    0m0.011s
sys     0m0.005s
```

Input:100

```
real    0m0.020s
user    0m0.010s
sys     0m0.009s
```

input:1000

```
real    0m0.367s
user    0m0.257s
sys     0m0.100s
```



The gale-shapley algorithm finds stable matchings with the asymptotic complexity of $O(n^2)$. During each iteration of the loop, there are n knights and each knight has n possible proposals, so the worst case scenario of this loop is n^2 .

The result does not exactly look like a quadratic curve. I believe this is because the time command includes reading the file and creating the dictionaries. With larger values of n , my graph would look closer to the graph of n^2 .